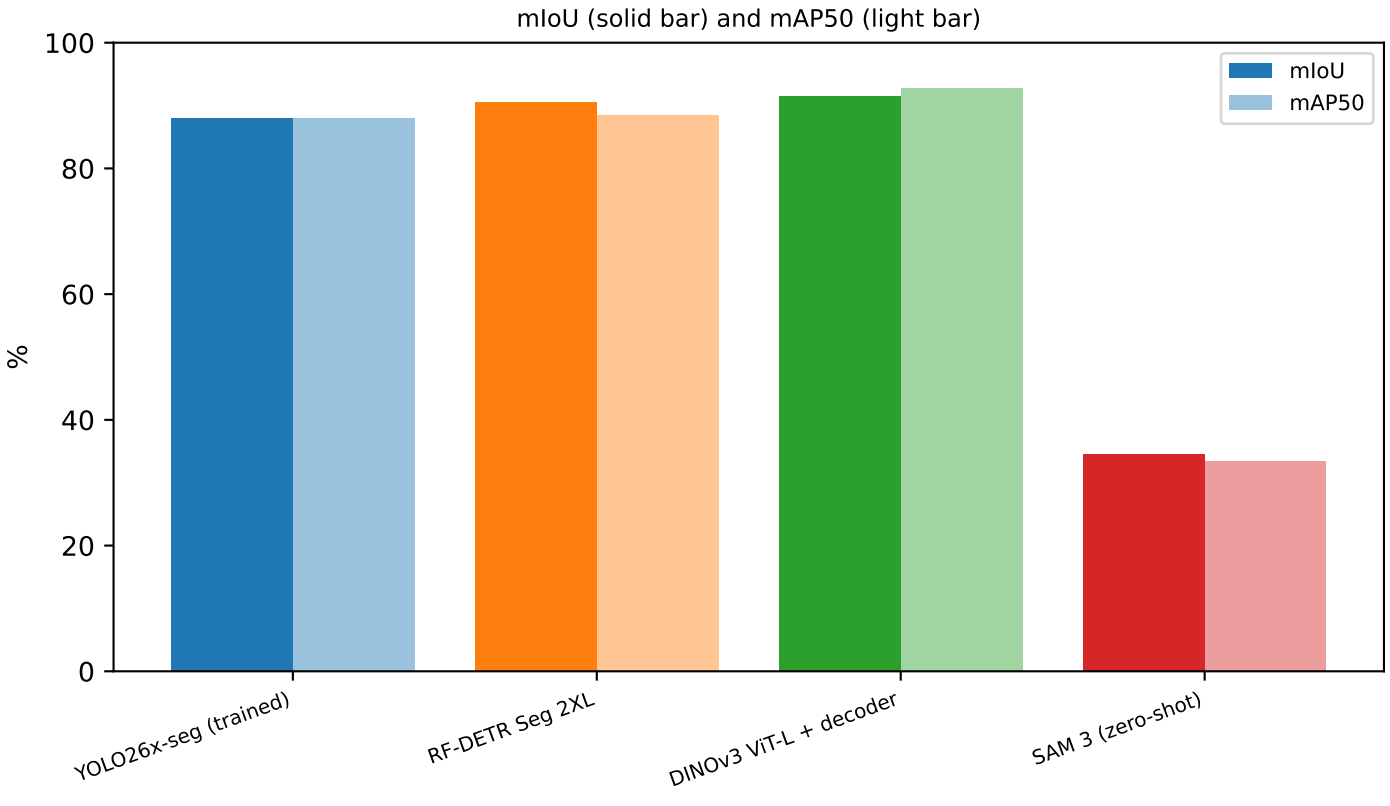
Comparison of car-part segmentation models

VehicleSeg10K dataset · test = 'valid' split (3069 images) · 2026-11-12

Best model by mIoU: DINOv3 ViT-L + decoder (91.5 %).

Confidence threshold for maps and detection: 0.25. A part counts as found if IoU ≥ 0.5.

Model	mIoU	mAcc	mDice	mAP50	mAP50-95	Part F1	ms/frame
YOLO26x-seg (trained)	87.9	95.5	93.5	88.0	76.2	92.0	38
RF-DETR Seg 2XL	90.6	94.4	95.0	88.5	73.9	91.2	77
DINOv3 ViT-L + decoder	91.5	95.3	95.5	92.7	79.8	93.2	1228
SAM 3 (zero-shot)	34.6	50.4	42.4	33.5	24.1	36.0	1342



Hardware, time per frame and memory

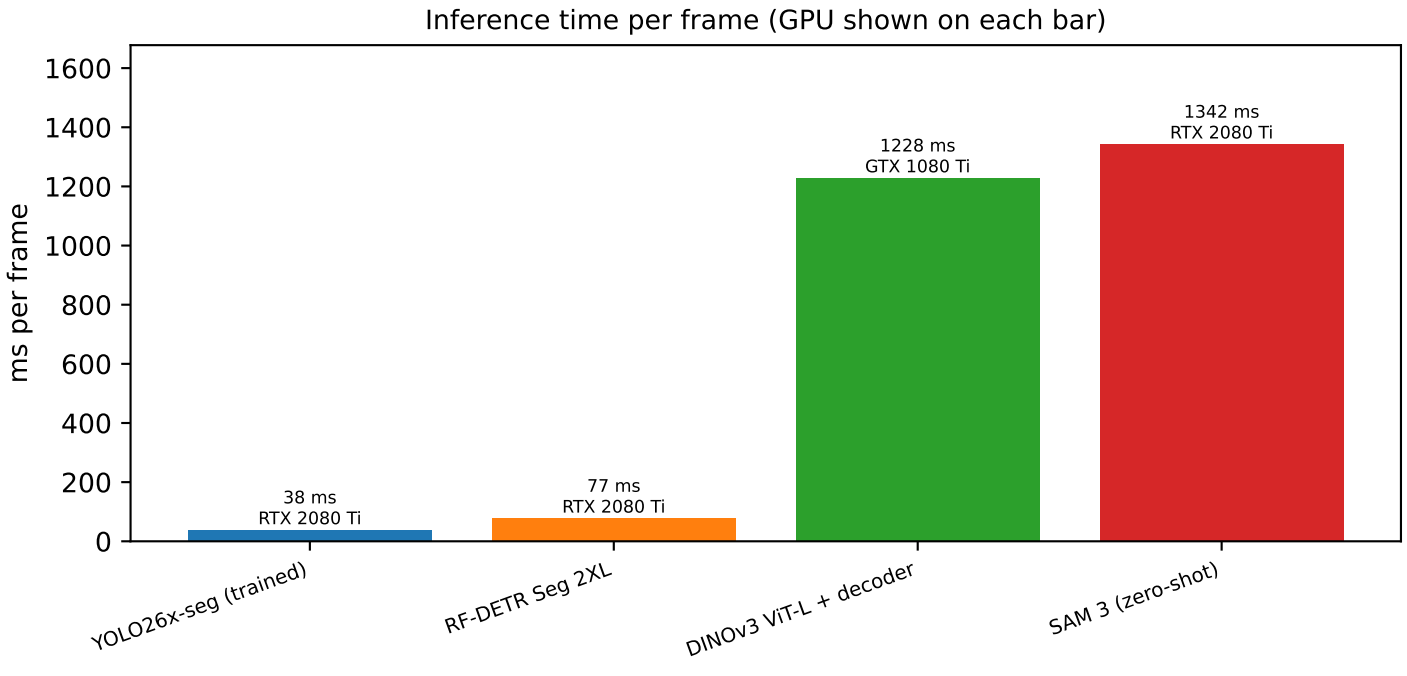
Servers. Two nodes with 4 GPUs each, sharing the same storage: one with 4× NVIDIA GeForce RTX 2080 Ti (11 GB, Turing architecture) and one with 4× NVIDIA GeForce GTX 1080 Ti (11 GB, Pascal). Neither has hardware bfloat16, so half precision means float16.

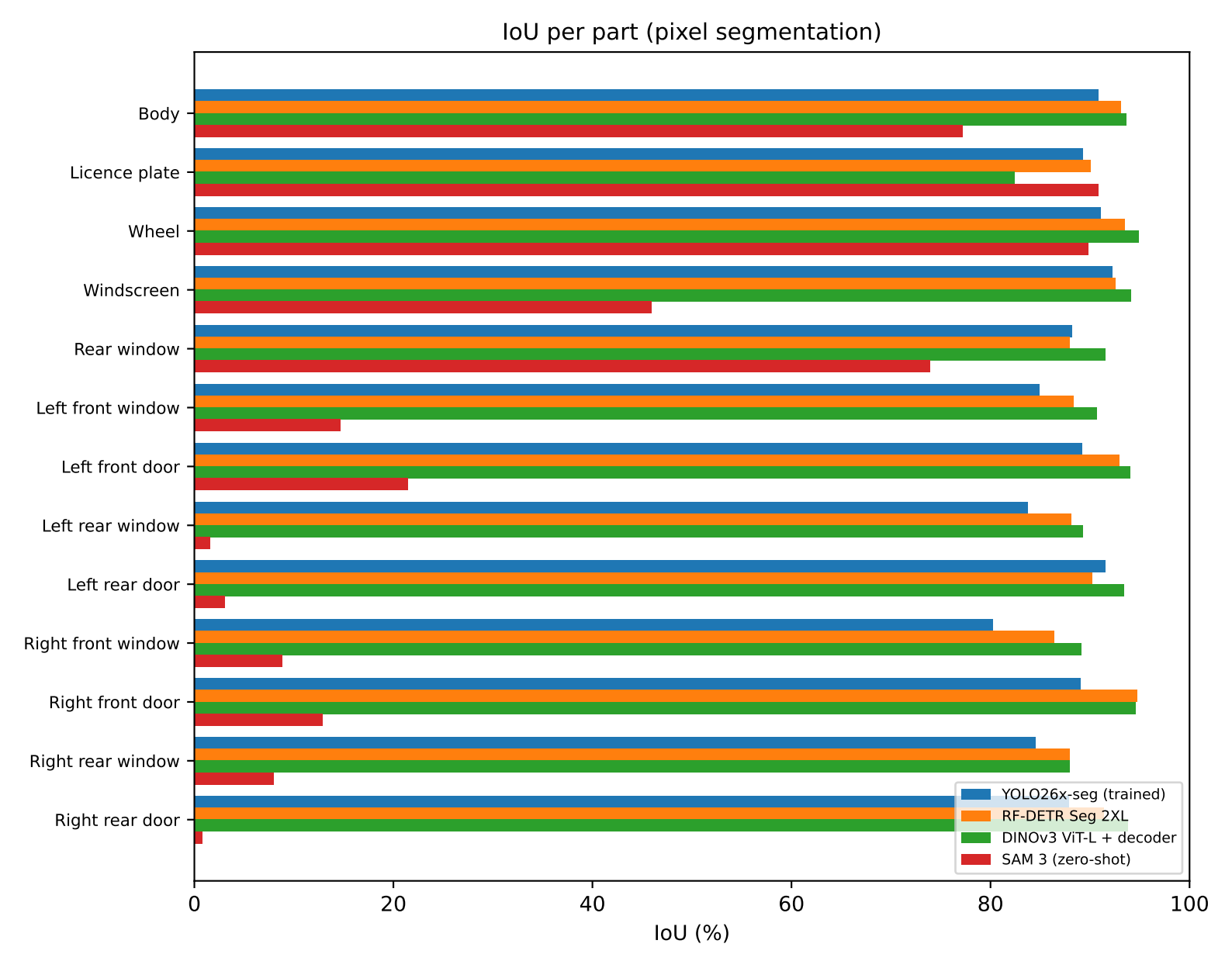
Time per frame. Measured on a single GPU with batch size 1, from the image already decoded in memory to the final masks: it includes pre-processing, the network and post-processing, but not reading from disk. The first image is discarded (warm-up). Test photos are large (up to ~12 Mpx) and masks are returned at full resolution, so post-processing matters. SAM 3 runs one text query per part (13 per photo). When inference is split across several GPUs, the time is per GPU, not the total divided.

Memory. 'Peak VRAM' is the maximum GPU memory reserved by PyTorch during the whole inference run (model + activations); it is the minimum the GPU needs. 'fp32 size' is the parameter count times 4 bytes. '-' means those predictions were generated before this was measured.

Note: speed of models run on the GTX 1080 Ti cannot be compared directly with those run on the RTX 2080 Ti: the 1080 Ti is slower and, having no tensor cores, runs float16 much more slowly.

Model	Trained on	Inference on	ms/frame mean	ms/frame median	FPS	Parameters (M)	fp32 size (MB)	Peak VRAM (GB)
YOLO26x-seg (trained)	4× RTX 2080 Ti	RTX 2080 Ti	38	35	26.4	71	269	2.6
RF-DETR Seg 2XL	4× RTX 2080 Ti	RTX 2080 Ti	77	68	13.0	38	146	7.4
DINOv3 ViT-L + decoder	4× GTX 1080 Ti	GTX 1080 Ti	1228	1147	0.8	-	-	-
SAM 3 (zero-shot)	not trained	RTX 2080 Ti	1342	1178	0.7	-	-	-




IoU per part (pixel segmentation)
Body
Licence plate
Wheel
Windscreen
Rear window
Left front window
Left front door
Left rear window
Left rear door
Right front window
Right front door
Right rear window
Right rear door
YOLO26x-seg (trained)
RF-DETR Seg 2XL
DINOv3 ViT-L + decoder
SAM 3 (zero-shot)
0
20
40
60
80
100
IoU (%)

IoU per part (%)

Part	YOLO26x-seg (trained)	RF-DETR Seg 2XL	DINOv3 ViT-L + decoder	SAM 3 (zero-shot)
Body	90.9	93.1	93.6	77.2
Licence plate	89.3	90.1	82.4	90.8
Wheel	91.1	93.5	94.9	89.8
Windscreen	92.2	92.6	94.1	45.9
Rear window	88.2	88.0	91.6	73.9
Left front window	84.9	88.3	90.7	14.7
Left front door	89.2	92.9	94.1	21.4
Left rear window	83.8	88.1	89.3	1.6
Left rear door	91.5	90.2	93.4	3.1
Right front window	80.3	86.4	89.1	8.8
Right front door	89.0	94.8	94.6	12.9
Right rear window	84.5	88.0	88.0	8.0
Right rear door	87.9	91.4	93.8	0.8
MEAN	87.9	90.6	91.5	34.6

Detection F1 per part (%)

Part	YOLO26x-seg (trained)	RF-DETR Seg 2XL	DINOv3 ViT-L + decoder	SAM 3 (zero-shot)
Body	96.4	77.2	98.8	76.6
Licence plate	90.9	93.1	91.6	91.8
Wheel	91.1	93.9	93.7	90.9
Windscreen	93.4	94.0	94.5	61.2
Rear window	88.6	90.9	91.7	55.0
Left front window	91.4	90.1	92.9	19.8
Left front door	93.6	93.4	93.4	18.5
Left rear window	92.2	93.3	93.8	21.9
Left rear door	92.3	92.5	91.7	0.5
Right front window	91.4	90.5	93.6	3.3
Right front door	92.6	93.1	92.6	5.9
Right rear window	91.3	93.1	93.6	16.7
Right rear door	91.3	91.1	90.1	6.0
MEAN	92.0	91.2	93.2	36.0

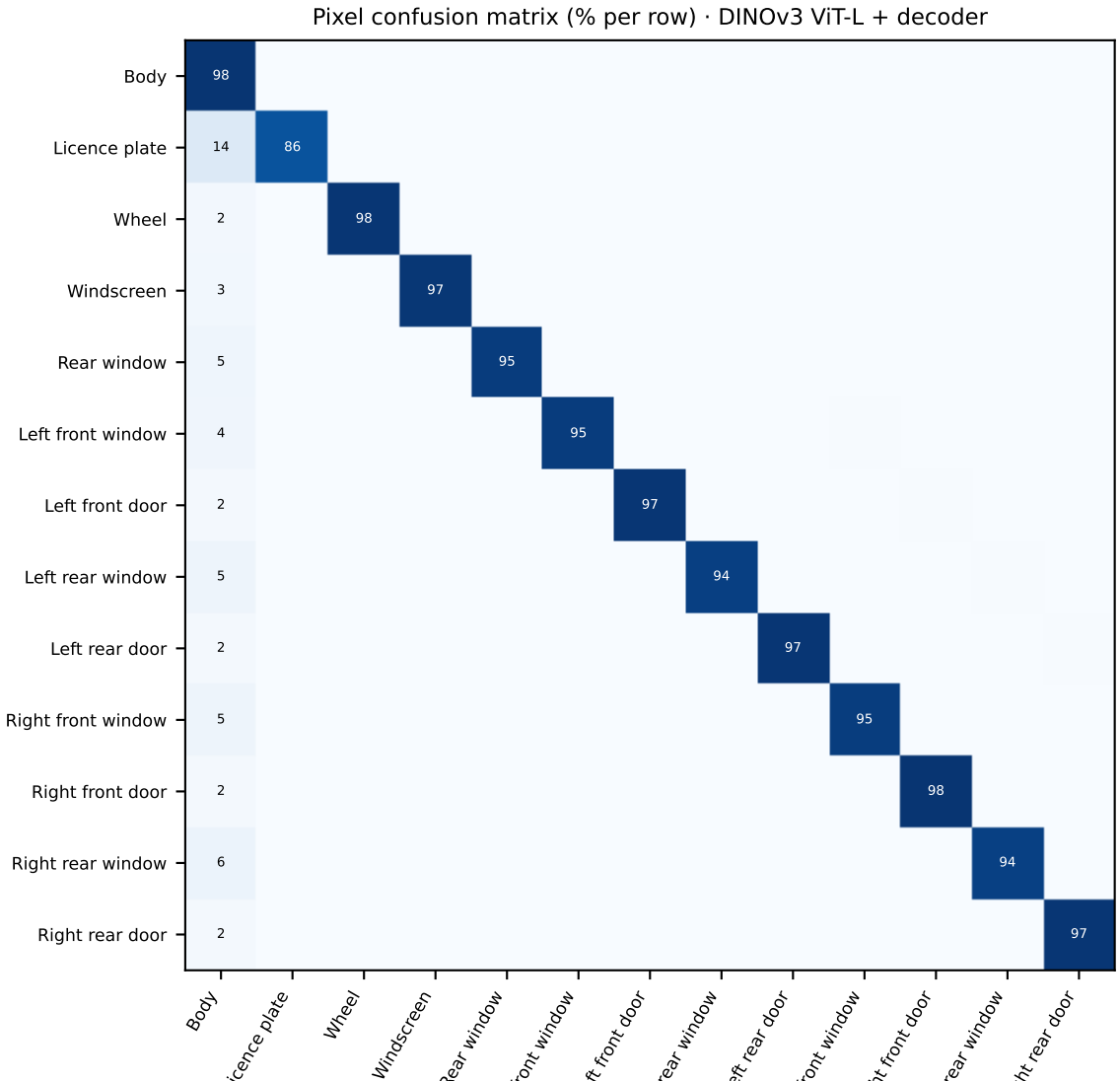
Detection recall per part (%)

Part	YOLO26x-seg (trained)	RF-DETR Seg 2XL	DINOv3 ViT-L + decoder	SAM 3 (zero-shot)
Body	93.9	77.1	99.1	76.7
Licence plate	86.3	91.3	91.9	96.3
Wheel	89.5	94.2	94.9	93.7
Windscreen	91.9	97.1	95.0	93.0
Rear window	84.5	90.7	95.2	90.3
Left front window	89.8	90.9	93.7	28.7
Left front door	91.9	93.1	93.2	24.1
Left rear window	91.1	92.2	94.8	32.2
Left rear door	90.3	91.9	92.8	0.8
Right front window	90.8	92.1	94.7	5.5
Right front door	91.3	92.0	93.6	9.4
Right rear window	90.9	93.5	95.8	30.0
Right rear door	89.7	91.0	91.8	9.7
MEAN	90.2	91.3	94.4	45.4

Left/right and front/back confusion

% of pixels of each pair of parts that the model assigns to the opposite part. Lower is better.

Pair	YOLO26x-seg (trained)	RF-DETR Seg 2XL	DINOv3 ViT-L + decoder	SAM 3 (zero-shot)
Left front window ↔ Right front window	0.35	0.72	0.34	21.07
Left front door ↔ Right front door	0.30	0.30	0.30	34.01
Left rear window ↔ Right rear window	0.32	0.40	0.30	16.21
Left rear door ↔ Right rear door	0.44	0.46	0.37	15.51
Windscreen ↔ Rear window	0.11	0.46	0.09	2.15




Examples (random test images)
Image
Ground truth
YOLO26x-seg (trained)
RF-DETR Seg 2XL
DINOv3 ViT-L + decoder
SAM 3 (zero-shot)

Methodology and caveats

Data. VehicleSeg10K: 8,596 training images and 3,069 'valid' images, with 13 classes: body, licence plate, wheel, windscreen, rear window and the four doors and four side windows with their side. 'Left' is the car's left (driver's point of view), not the left of the photo. All models are trained on 'train' and evaluated on 'valid', which is not used to pick a checkpoint (the last one is evaluated).

Pixel metrics. For each image a label map is built: first the body and the parts on top, the most confident one last. It is compared with the official PNG. mIoU and mAcc are computed over the 13 classes without background, accumulating pixels over the whole test set.

mAP. With one mask per class and image (the union of the model's instances with confidence ≥ 0.25) against one ground-truth mask per class and image, using pycocotools.

YOLO. Ultralytics YOLO26-seg from COCO weights, without horizontal flipping (a flip swaps left and right and would corrupt the labels).

RF-DETR. Roboflow RF-DETR Seg 2XL from its pre-trained weights, 50 epochs, also without flipping. The EMA weights of the last epoch are evaluated.

DINOv3. Meta's ViT-L/16 backbone, frozen, with an FPN-style decoder trained on top (40 epochs). It is pure semantic segmentation: it outputs a per-pixel map, not instances.

SAM 3. Meta's text-prompted model, with one query per part (e.g. 'left front door'). Zero-shot uses it as released, without training on this dataset; the fine-tuned version is trained with the official trainer and a frozen backbone.